FOR K IN THE DERIVED CATEGORY OF THE SMALL ETALE TOPOS OF A SCHEME S THE ADJUNCTION K -> R PI_* PI^{-1} K IS AN ISOMORPHISM

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The counted proof is the author's complete argument: it identifies the exactness of the two functors and the fact that their composition is the identity as the whole reason, which is exactly the crux.

Editorial note. The label lemma-cohomological-descent-etale defines the statement; the counted proof is the proof environment immediately following the lemma in the source.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217). The class block and the externaldocument lines of the native preamble are not included, because the assembler supplies the wrapper class and those external documents are not part of this material.

Allowed standard prerequisites (editorial).

- The morphism pi : S_etale -> S_{etale,big} and its pushforward. Same chapter, the morphism of topoi pi_S : Sh(S_etale) -> Sh(S_{etale,proetale}) / big etale topos, and the identification pi_{S,*} = i_S^{-1} used in the proof. Supplies the exactness of pi_S^{-1} and pi_{S,*} which is exactly the content of the one-line argument.

Lemma 0.1. *Let S be a scheme. For $K \in D(S_{\acute{e}tale})$ the map*

$$K \longrightarrow R\pi_{S,*}\pi_S^{-1}K$$

is an isomorphism.

Proof. This is true because both π_S^{-1} and $\pi_{S,*} = i_S^{-1}$ are exact functors and the composition $\pi_{S,*} \circ \pi_S^{-1}$ is the identity functor. □

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